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# Advanced Training and Simulation Hand Grenades: A Comprehensive Overview

Modern military and law enforcement training increasingly employs specialized hand grenade simulators to safely and realistically emulate live grenades. These training grenades range from high-tech smart devices with electronic scoring to simple inert shells with pyrotechnic charges. Below, we verify and elaborate on notable examples, delve into underlying technologies (like proximity sensors and MILES integration), and compare their use cases, costs, and effectiveness in various training scenarios.

## Sim Grenade™ (Serious Simulations) – Features and Applications

**Technical Features:** The **Sim Grenade™** (developed by Serious Simulations, USA) is an intelligent, reusable grenade simulator with an internal "electronic brain" and **patent-pending proximity sensors**. Upon detonation, it can **measure the exact distance to human targets within its blast radius** and determine their wound/kill status in real time [1] [2]. This is achieved regardless of where or when it was thrown, solving a long-standing training gap where fake grenades relied on subjective adjudication. The Sim Grenade transmits these results wirelessly to exercise controllers and the affected individuals, providing immediate feedback [1]. It recreates key functional elements of a real M67 grenade: users must pull a pin and release a spoon (lever), after which a ~5-second delay fuse runs before "detonation" [3]. The device is carefully weighted and balanced so as not to alter the **throwing range or feel** compared to a live grenade [3]. Critically, the Sim Grenade produces a **visible flash and audible bang** when it "explodes" (sufficient for all participants nearby to see and hear), yet uses no high explosives and does not project dangerous fragments [4]. Its outer casing includes protective padding, allowing it to be safely thrown at or near people without injury [5] [6]. The unit is **rugged and fully reusable**, designed to be easily reassembled and serviced for repeated training use [7].

**Applications in Training:** The Sim Grenade was created to enable **realistic force-on-force training** with grenades, a capability previously considered impractical or unsafe [8] [9]. Traditional training often resorted to throwing dummy objects (tennis balls, rocks, etc.) to simulate grenades, with instructors subjectively calling out hits [8]. The Sim Grenade fills this gap by allowing **maneuver units to train with grenades under realistic conditions** – for example, during **platoon or company-level combined arms exercises**, urban combat training (MOUT/CQB), or other scenarios where participants actively employ grenades against opposing force role-players. Because the device automatically **assesses casualties based on actual proximity and intervening cover**, it removes the need for "honor system" kill adjudication and prevents negative training (e.g. players ignoring grenade effects) [1] [2]. This enables objective scoring and After-Action Reviews (AARs): observers can replay who was where and whether they were in effective cover when the grenade went off. In practical terms, the Sim Grenade **greatly enhances realism and accountability** in **force-on-force exercises** – soldiers learn to react properly to grenades (taking cover, etc.) knowing the device will **"kill" them if they do not**. It is also useful in **urban combat and CQB training**; while its flash/bang is not as overpowering as a real flashbang, it does add sensory stress and forces quick decision-making under simulated combat conditions. Because it can distinguish adequate cover (e.g. a thick wall or ditch) from flimsy cover (bushes, thin walls) and only "kill" those truly exposed [7], [10], it reinforces correct tactical behavior. Overall, Sim Grenade's combination of realism (functional and ballistic) and safety makes it ideal for

**high-end training environments** like dedicated combat training centers, special forces urban warfare facilities, or advanced live exercises that can invest in the needed instrumentation.

**Verified Details:** The Sim Grenade's advertised capabilities have been confirmed by official sources. A UCF Business Incubation Program release (Serious Simulations is a graduate) states that the device determines proximity to targets on detonation and immediately communicates wound/kill status to the individuals and exercise managers [1]. It also explicitly lists the features mimicking a real grenade (pin pull, spoon fly-off, 5-second delay, correct weight/throw, audible/visible detonation, realistic response to cover, and reusability) [11] [4]. Additionally, a 2025 Orlando news article notes the Sim Grenade uses technology to calculate exact distances to soldiers/vehicles for real-time battle damage assessment, and that it **replicates the effect of cover** (penetration of intervening materials) to give accurate feedback on protective cover usage [12] [10]. The same source highlights the inclusion of a **realistic spoon, pull ring, and safety clip** on the grenade, with a padded exterior for safe use [13]. These details corroborate the technical claims and indicate the product is now transitioning to production for the U.S. Army's Synthetic Training Environment program [14] [15].

**In summary,** the Sim Grenade stands out for bringing **true casualty simulation** and **data connectivity** to grenade training. It is best suited for advanced force-on-force exercises and scenarios requiring **detailed performance tracking**, albeit with the trade-offs of higher unit cost and reliance on electronic systems.

## Proximity Sensor Technology and MILES/STE-LTS Integration (Sim Grenade)

One of the most innovative aspects of the Sim Grenade is its integration into the **MILES** (Multiple Integrated Laser Engagement System) ecosystem and the Army's next-gen **STE-LTS** (Synthetic Training Environment – Live Training System). The grenade uses a **wireless signal and sensor network** to register hits on players just like MILES gear for rifles/tanks does, but without needing the traditional laser "beam" from the grenade. Instead, the Sim Grenade's internal sensors and processing can determine distances to targets and whether those targets are behind cover. How exactly does it achieve this? While exact proprietary details are not publicly disclosed, the system likely uses a combination of **RF signaling** and **timing or power measurement** (or potentially ultrasonic or IR pulses) to gauge distance to each instrumented player. Each soldier in the exercise wears a **Universal Receiver Unit (URU)** as part of their personal MILES-like gear [16] [17]. When the Sim Grenade detonates, it emits a coded signal received by all URUs in the vicinity. By measuring signal characteristics (such as time of flight, strength, or coded distance thresholds), the URU can determine how close the grenade was and whether the soldier is within the casualty radius. The URU then processes this and can either directly indicate a hit (e.g. by sounding a tone on the vest or disabling the soldier's laser transmitter) and/or send the data up to the exercise control system [16] [18].

According to PEO STRI, the **M67 Grenade Training System** (which includes Serious Simulations' device) interfaces with a **Universal Receiver Unit securely fastened to the soldier's gear** [16] [19]. The URU processes signals from the sim grenade during detonation to determine effects. The system is explicitly designed to interoperate with **MILES**, meaning it can tie into legacy laser engagement frameworks [20]. In practice, this ensures that when the Sim Grenade "kills" or "wounds" someone, it is recognized by the same instrumentation that tracks laser tag hits, allowing seamless integration of grenade effects into overall exercise results. The Serious Simulations device was developed under the Army's STE-LTS program (with support from entities like PEO STRI and PM TRADE), which focuses on filling live training gaps such as thrown and placed munition effects [21] [22]. The grenade's data-rich output can update

**scoreboards and AAR systems in real time** – for example, it can log a kill in the exercise network immediately [15] [23].

Beyond just "hit/miss," the Sim Grenade's sensors account for **cover and environment**. The device is smart enough to know if a target was behind adequate cover and thus should be unscathed [7]. It likely does this by line-of-sight considerations: perhaps using multiple receivers on the URU to infer whether a direct path existed or by attenuation of the signal. In any case, this advanced sensing prevents players from being "unfairly" killed through walls or, conversely, avoiding effects they should have suffered.

**Proximity Sensor Function:** In effect, the grenade creates a volume of effect (comparable to a real grenade's lethal radius) and any URU within that volume, not shielded by substantial cover, will register a hit. The accuracy is such that it can gradate wounds vs. kills by distance – e.g. someone at 3 meters might be "wounded" while someone at 1 meter is "killed," mirroring real frag pattern lethality. This is a major leap from older training simulators that often used fixed laser beams or had no way to distinguish distances.

**Data Transmission:** The data link from grenade to URU is wireless and presumably low-power and short-range. It might use a specific RF band (possibly similar to MILES lasers but radio-frequency, or a coded IR omnidirectional pulse). PEO STRI refers to it generally as "signals emitted by the sim grenade" processed by the URU [16] [18]. The **STE-LTS** program ensures this is integrated with the overall **training system architecture**, meaning the results can be fed into exercise control software and **after-action reviews** seamlessly [15]. Indeed, a news release confirms that the Sim Grenade **reports its damage results in real time and is compatible with legacy systems** (like existing MILES gear) [24]. In summary, the Sim Grenade's proximity sensing and MILES integration represent a cutting-edge application of IoT-like sensors and real-time telemetry in live training.

By combining precise sensing and networked data, the device enables **detailed performance analytics**: trainers can later review exactly who was in the casualty radius and whether their positioning (prone vs standing, behind cover, etc.) saved them. This level of fidelity was previously only possible in instrumented virtual simulations or with labor-intensive post-exercise analysis; now it can occur live on the field. The **technology effectively links physical actions to digital feedback**, enhancing realism without compromising safety.

**Example:** During a platoon **force-on-force exercise**, a squad tosses a Sim Grenade into a room. Two opposing soldiers are inside: one dives behind a sturdy concrete pillar, the other is caught in the open. The grenade detonates – both hear the bang and see the flash. Instantly, their URUs receive the grenade's signal. The soldier behind the pillar has the signal blocked sufficiently (or the system recognizes adequate cover) – his URU may register a "near miss" or no casualty. The exposed soldier's URU, however, registers a lethal effect. His MILES gear could emit a kill tone and deactivate his weapon transmitter. Controllers monitoring the exercise see the status update. Such granular, **real-time kill notification** greatly streamlines exercise management and **eliminates ambiguity** ("am I hit or not?") for participants [24].

Overall, the Sim Grenade's integration with proximity sensors and MILES/STE-LTS exemplifies the **modern trend of connected training systems**, where even grenades become part of the digital network. This yields more **objective, data-driven training** while preserving the physicality of live exercises.

## Nammo Training Hand Grenade – Reusable Fuse Mechanism and NATO Adoption

**Technical Characteristics:** The **Nammo Training Hand Grenade** (developed in Norway/Finland by Nammo) is a **realistic inert grenade** designed for basic throwing and handling training. It is built using many of the same parts as a live grenade (including a real grenade body shell in some cases) but filled with an inert substance instead of high explosive [25]. **Weight, size, and balance** closely match those of tactical (live) grenades, ensuring soldiers build muscle memory and correct handling techniques [25]. Critically, the training grenade uses a specialized **reusable fuze system** that mirrors a live fuze's delay and function but with a small pyrotechnic output. The **training fuze assembly** incorporates a **3–5 second delay element (just like a live M213 or M228 fuze) and a tiny pyrotechnic charge** [26]. When the trainee pulls the pin and throws the grenade, the fuze ignites and after the delay emits an **audible "pop" or report** to signal the moment of "explosion" [27] [26]. This sound effect is loud enough for the user and instructors to hear clearly (simulating the grenade's bang) but is much smaller than a real grenade's blast and **does not produce dangerous fragmentation**. In fact, the grenade's body often remains intact or only comes apart in a safe way so it **can be recovered and reused**. Nammo's design allows the **body to be used multiple times** – after each throw, one can replace the spent fuze with a fresh training fuze for the next exercise [28]. The metal or composite body can withstand repeated use, making it a cost-effective solution after the initial purchase.

The Nammo training grenade is **safety-certified for use in limited-area ranges**, meaning it can be thrown in relatively confined training areas without risk (no shrapnel and only a small smoke/bang) [29] [30]. This is important for armies that may conduct grenade practice in smaller facilities or for police/military units that want to practice throws outside of dedicated demolition ranges. According to Nammo, these grenades are *"safe for use on size-restricted training ranges"* [28] [30], implying that their hazard zone is minimal (likely just a few meters for the small blast noise). They come packaged like tactical grenades and include user manuals on safe use [26].

**Reusable Fuze Mechanism:** The core of the reusability is the **training fuze**. It is built with all **production parts of a real fuze** (so the feel and function – pin, striker, lever, delay powder train – are identical) [26]. However, instead of a detonator that would set off high explosive, it has a tiny charge (pyrotechnic flash) just to make noise. After functioning, the expended fuze can be unscrewed and discarded. A new training fuze is then screwed into the grenade body for the next trainee or throw. This modular approach significantly cuts down long-term costs: rather than using up an entire grenade each throw, one only consumes the **fuze unit**. The heavy steel body can be reused dozens if not hundreds of times (subject to inspection for cracks). Nammo's product description confirms *"these modules are reusable multiple times by replacing the training fuze assembly"* [26]. It essentially bridges the gap between completely inert dummy grenades (which have no noise or function) and one-time simulators – providing both realism and reusability.

**Adoption in NATO Forces:** Nammo's training grenades have seen adoption in the Nordic countries and are making inroads in NATO. They were **qualified by the Finnish Defence Forces** as the standard training grenade for Finland [30] [31]. In fact, the grenade has NSN (NATO Stock Numbers) assigned under Finland's country code, indicating it's an approved item in their inventory [30]. Finland's use implies the grenade meets stringent safety and performance standards – Finnish conscripts regularly train with them instead of live grenades until final qualification. Given Nammo's Norwegian-Finnish roots, it's likely also in use (or being evaluated) by Norway (a NATO member), though the source explicitly mentions Finnish qualification. The concept of **reusable training fuzes** is not entirely new – some Western armies historically had practice grenades with small black powder charges – but Nammo's version is modernized and has gained attention. A 2024 defense article highlighted that the

U.S. had early fielding of Nammo's **Scalable Offensive Hand Grenade (SOHG)** and emphasized the importance of training versions for any new grenade [32]. This suggests the U.S. is aware of or using Nammo's technology for training (especially since the SOHG offensive grenade would need a training variant). Indeed, recent news indicates the U.S. Army is procuring a **Scalable Offensive Grenade** from Nammo (designated MK21 Mod 0) and that implies a training grenade package would accompany it [33] [34]. Additionally, other NATO countries that have tight environmental regulations, like Germany or the Netherlands, may be interested in low-hazard training grenades – though specific public info is limited, **Nammo's grenade is marketed internationally as a NATO-compatible item**.

**Use Case:** The Nammo training grenade is **ideal for "Level 1" grenade training** – teaching recruits how to properly handle and throw grenades. Its **realistic weight and components** build correct habits: holding the grenade safely, pulling the pin with the proper technique (while keeping the lever secure), employing proper throwing stance, and then taking cover. The small report conditions new soldiers to anticipate the bang (building confidence and timing without the full danger of live grenades). It's also useful for **refresher training** and competitions – e.g., practicing throwing accuracy at targets on a range. Because the bodies are durable, units can practice repeatedly at moderate cost. For safety, these grenades still require some precautions (ear protection is usually recommended due to the sound, and a minimal throwing distance so the blast is not right next to the ear or body). However, unlike live or larger simulators, they can be used in scenarios with instructors and other trainees in fairly close proximity, even scenarios like throwing grenades near bunkers with fellow soldiers nearby (to simulate battlefield conditions of tossing grenades over walls or through windows while a teammate is covering). The **limitation** is that Nammo's training grenades **do not have any electronic scoring or sensor** – they do not "know" if someone was within radius. They are strictly to train the **act and mechanics of grenade throwing**, not to simulate its effects on targets. Thus, for force-on-force games, they aren't useful unless augmented by instructors or coupled with MILES lasers (some armies have attached MILES grenade simulators to such bodies, which is another approach).

**Summary:** Nammo's reusable training grenade demonstrates a **cost-efficient, widely applicable solution** for grenade handling training. It provides **realism in form and function** (actual delay fuze with pop sound) while being **safe and re-usable**. Many NATO forces favor such designs because they reduce range restrictions and cleanup (just pick up the bodies and spent fuzes) and **lower long-term costs**. Finnish adoption validates its reliability. These grenades fill the need for routine training (every soldier learning to throw properly) before one graduates to either live fragmentation grenades or more sophisticated simulators like the Sim Grenade for tactical exercises.

## FBSS-Mini (N589) – Wescom Defence's Flash-Bang-Smoke Simulator

**Technical Features:** The **FBSS-Mini** (Flash-Bang-Smoke Simulator, product code N589) by Wescom Defence (UK/Germany) is essentially a **small pyrotechnic insert** that can be placed into a dummy grenade body to simulate a realistic grenade blast. In other words, it's a **"pyro core"** that, when ignited, produces a **bright flash, a loud bang, and a cloud of white smoke** – mirroring the visual/auditory effect of a grenade without any fragment hazard [35]. The unit is quite compact (on the order of a few centimeters) and contains a small flash charge (approximately 0.8 grams of flash powder) [36]. On activation (usually triggered by the striker mechanism of the training grenade body it's housed in), the FBSS-Mini yields an **extremely loud report of about 167 dB at 10 cm distance** [37]. This is roughly equivalent to the sound of a real grenade or flashbang at very close range, providing an **overwhelming bang and blinding flash** to anyone nearby. It also generates a burst of **thick smoke** (white smoke from the pyrotechnic composition) to mimic the dust and smoke of an explosion [38]. Importantly, it produces **no shrapnel** – the device's casing is designed to either remain intact or break into harmless pieces.

Typically, these simulators are used by inserting them into **inert grenade bodies** made for training (for example, Saab or other manufacturers have dummy grenade shells with cavities for detonators or simulators). When the grenade's pin is pulled and the lever released, an initiation mechanism (primer) lights the FBSS-Mini after the delay, resulting in the flash-bang event.

Wescom's FBSS series is essentially the **same pyrotechnic effect used in military artillery simulators and flashbangs, miniaturized** for hand grenade form factor. The FBSS-Mini's sound level of **167 dB at 0.1 m** implies that at a typical exposure distance (say a few meters) it's still extremely loud (likely >130 dB) – definitely **ear-ringing and disorienting**, though users are generally required to wear ear protection during training with these devices [39]. The device's spec lists a safety distance of about 10 cm directly in front (mainly to avoid being right on top of it) with **ear and eye protection mandatory** [39]. In practice, it is safe to be in the same room when it goes off, but one should not be holding it in hand when it ignites. The flash is very bright (comparable to a camera flash but sustained and more intense, enough to momentarily blind in a dark room) and the smoke adds realism and some obscuration.

The FBSS-Mini is **used by the Swedish Army and other NATO forces** as a stress inoculation and simulation tool [40]. In Sweden, for instance, these simulators are employed in force-on-target exercises and urban combat training to get soldiers accustomed to grenade blasts – the doctrine of *"stress inoculation"* means repeatedly exposing troops to loud noises and flashes so that they are not paralyzed by sensory shock in real combat [40]. Other NATO users likely include countries that have Saab's training systems, since Saab (a Swedish company) integrates Wescom pyrotechnics in their solutions. The **N589** designation suggests it's one of Wescom's catalog items (indeed described as a "Training grenade" on their site). The FBSS-Mini essentially provides a **higher-fidelity explosion effect** than the small pop of a Nammo grenade, albeit at the cost of being single-use and requiring pyro handling procedures.

**Application in Training:** The FBSS-Mini (and similar **flash-bang grenade simulators**) are particularly valuable for **CQB (Close-Quarters Battle) and MOUT** scenarios where the **sensory effect of a grenade** is important. For example, special forces or infantry clearing a building need to experience the reality of a grenade's flash and bang so they learn to keep their head down, cover ears, or use proper stun techniques. While devices like the Sim Grenade register hits, they do not fully replicate the **concussive feel** of an explosion. The FBSS-Mini, by producing a loud noise and flash, **induces realistic stress and momentary disorientation** – crucial for training troops to operate under battlefield conditions (e.g., not flinching or panicking when a grenade goes off nearby). It's especially useful in **room entry training, hostage rescue drills, trench clearing and other scenarios** where grenades are used at very close range to targets. Because it has no actual fragments, troops can throw it at very close distances or even "safely" near role-players (with protection) to simulate disabling a target with a concussion grenade, for instance. Law enforcement tactical teams (SWAT) could also use such simulators for practicing flashbang use in urban settings.

**Integration and Variants:** In many training systems, the FBSS-Mini can be combined with **laser-based simulators**. For instance, one might place an FBSS-Mini inside a laser-emitting training grenade body: upon detonation, the flash-bang goes off and simultaneously a MILES laser is triggered to "score" hits. This hybrid approach gives both a **physical effect and a scoring mechanism**. Some modern systems indeed allow that, as noted by sources that mention combining with MILES or other sensors to trigger a laser when it explodes [41] [42]. This way, you don't sacrifice objective hit assessment.

**Safety and Limitations:** Despite being far safer than live grenades, the FBSS-Mini **is still a pyrotechnic**. This brings several constraints: a dedicated **safety radius** (commonly, all trainees within a few meters must wear ear protection and eye protection), thorough **safety briefings**, and range control measures to prevent fires (the device can potentially start small fires, though it mostly just flashes and

smokes). Instructors must also account for the smoke in indoor facilities (ventilation needed). There is also **additional cost** – each activation burns one unit, so for every throw you incur the cost of a pyrotechnic cartridge (which, while cheaper than a live grenade, is more expensive than a reusable system's per-throw cost). Another limitation is **cleanup**: after many uses, the area will have scorch marks and fragments of the simulator casing or lots of residual smoke. But these are manageable with proper planning.

In summary, the FBSS-Mini (N589) is **the go-to solution for realistic grenade blast simulation** in training. Its **strength is creating the sensory and stress effects** of a real grenade explosion (flash, bang, smoke) very faithfully [35]. This makes it invaluable for **combat stress training, urban warfare drills, and any scenario where troops need to react under pressure**. The **trade-offs** are that it doesn't inherently provide hit scoring and requires pyrotechnic handling and one-time use expenditure. Thus, it is best **combined with other systems**: for instance, in a force-on-force exercise, one might use a Sim Grenade for scoring most of the time, but occasionally also trigger an FBSS-Mini in a controlled way to keep soldiers on their toes with a loud blast. Indeed, some Swedish exercises reportedly use them during live maneuver training to simulate incoming grenade or artillery blasts, forcing troops to practice immediate action drills under duress.

## SOHNI THG67 – Biodegradable Body Training Grenade and Close-Range Safety

The **SOHNI THG67** is an example of a **"low-risk" practice grenade** that prioritizes safety and environmental friendliness. (The name *SOHNI* THG67 appears to refer to a particular model or manufacturer; one source suggests it's an Estonian-made grenade often simply called THG67.) This class of training grenade is characterized by a **biodegradable or frangible body** and an extremely low hazard pyrotechnic, if any. Essentially, the grenade's body is made of **non-metallic, easily crumbling materials** – often a mix of **naturally degradable substances like clay, compressed soil, or soft polymer** that will break apart without sharp fragments [43] [44]. For example, one variant uses a body of pressed clay and a fuse head of bio-plastic [43]. When such a grenade detonates (some are purely inert with a small pop, others might have a mild flash charge), the body either remains intact or shatters into **soft, harmless pieces**. There are **no metal fragments, springs, or hard shards** to injure anyone – even if a soldier detonates it right next to themselves (accidentally or in "buddy throw" drills), it should not cause injury beyond perhaps a slight burn or ringing ears. One manufacturer's description emphasizes that **the fragmentation and flame from the explosion do *not* cause injury, even with a very close explosion** [43]. This indicates how low-power the charge is – just enough to burst the body and make noise, but not enough to harm exposed skin at close range.

**Biodegradability:** A major feature is that these grenades are **environmentally friendly**. Materials like clay and bio-plastics will **naturally degrade outdoors** over time [43]. This addresses a common issue with frequent training: traditional practice grenades litter ranges with plastic caps, metal pin rings, etc. Biodegradable grenades leave minimal long-term debris – any remnants disintegrate with rain or sunlight. For militaries that train in forests, grasslands, or near public land, this is important to meet environmental regulations and avoid costly cleanup. An **Army Technology** whitepaper on an eco-grenade notes it's made of degradable materials and explicitly says it's designed for **environmental protection** [43]. It also cites that it provides realistic effects (sound, smoke, body blast) but with great safety such that soldiers gain confidence handling live grenades without risk [43].

**Use in Close-Range Training:** Because of their safety, grenades like THG67 are ideal for **introductory training and mass training events**. New recruits can practice throwing them *in very close proximity to others*. For instance, there are classic drills where a soldier throws a grenade while another soldier

stands nearby (to simulate throwing from a trench with a teammate right behind you, or throwing over a wall while others provide covering fire). With live grenades this is extremely dangerous; even with standard practice grenades, you wouldn't throw near someone. But with a THG67 type, you **can safely conduct buddy-team throwing exercises** – if a grenade were to go off at a comrade's feet due to a mistake, it wouldn't cause serious harm. This allows training of scenarios like "grenade sumo" (practicing throwing a grenade past an obstacle where fellow soldiers are on the other side). Moreover, in recruit training, instructors can have soldiers throw these grenades in relatively small pits or ranges without heavy protective works, since even an accidental close detonation won't be deadly.

**Absolute Safety Features:** These grenades often have either **completely inert fuzes or a very small charge**. Some versions might just have a primer that makes a "pop" and maybe emits a puff of smoke for visual cue. Others might include a *tiny* flash charge to give a bit of bang, but still much less than FBSS or even Nammo's training fuze. The focus is on **eliminating all injury potential**: one manufacturer advertises that even if it explodes right by your hand, it shouldn't break bones or lacerate – which is a bold but telling claim. Practically, users should still wear ear protection if there's any bang, and eye protection is always recommended. But compared to any other type, this is the safest. They are described as "risk-free" or "low-risk" grenades for training [45].

**Example – K417 and THG67:** The Korean company CNO Tech makes the **K417** eco-friendly grenade (and similar models CN69, etc.) which align with this concept. Their specs: body of PLA (biodegradable polymer) and clay, matching weight of real grenade, and goal of "utmost safety even if it explodes near the human body" [44] [46]. The THG67 from Europe is in the same vein; it has been marketed by a company (Bristol Trust) and trialed by some European forces. The Malaysian Army even evaluated it (per a Malaysian Defence article) for reducing training area cleanup. Europe's trend toward green training solutions has boosted these products.

**Training Value:** These grenades shine in the **very early stages of hand grenade training** – often termed Level 1 familiarization. They allow repetition of the basics: pulling pins correctly, feeling the fuze delay, throwing and taking cover with the right timing. The soldier gets a **psychological feel** for a grenade going off (sound/smoke, though mild) which prepares them for later using a live grenade. They are also great for scenarios like **platoon exercise in sensitive terrain** (say a training exercise on an "eco-friendly" base or island where you cannot leave metal debris – using biodegradable grenades means after a rain, there's virtually no trace).

**Limitations:** The obvious downside is the **lack of "excitement" or advanced features**. The sound and flash are minimal – nowhere near what a real grenade or even an FBSS simulator does. Thus, they do not effectively stress or test a soldier's reaction under battle-like conditions; they are meant for skill learning, not stress inoculation. There is **no built-in scoring or electronics** – they are entirely analog. If used in force-on-force, they cannot "tell" who is hit; they'd rely on umpires. However, they're generally not intended for force-on-force (except maybe as harmless props). Another limitation is that because they are so light on effects, a trainee might not develop a full respect for grenade lethality – so typically these are used to build confidence first, then the trainee graduates to something louder or eventually a live throw so they appreciate the real danger.

In short, the **SOHNI THG67 and similar biodegradable training grenades** are tailored for **large-scale, entry-level training and environmentally sensitive use**. They let armies train grenade handling to hundreds of soldiers safely, even on grass fields or wooded areas, without injury risk or cleanup issues. They sacrifice explosive realism for these benefits. They wouldn't be used to train combat scenarios with opposing forces (because they don't simulate casualty effects or loud blasts), but they ensure every soldier can competently throw a grenade before ever using a live one. As an analogy, if the Sim Grenade

is a high-tech **"Level 3" trainer** and the Nammo a **"Level 2" trainer**, then the THG67-type is **"Level 1"** – maximum safety, moderate realism, used early and widely.

## Enhanced Tactical Multi-Purpose (ET-MP) Grenade – Development Status and Timeline

The **ET-MP (Enhanced Tactical Multi-Purpose)** grenade is a *real* next-generation U.S. combat grenade (not a training device per se) under development by the U.S. Army's Armament Center at Picatinny Arsenal. It is included in this discussion to illustrate **technological trends (like electronic fuzes and multi-mode functions)** that will eventually influence training grenades. The ET-MP is designed to be the first new lethal hand grenade in decades for the U.S., with two primary novel features: **selectable blast modes** (fragmentation vs. concussion) and a fully **electronic fuze** for precise timing and improved safety [47] [48].

**Technical Features:** The grenade has an ambidextrous design (no separate versions for left-handed throwers) and a **switch that allows the user to choose the detonation effect** – either a fragmentation grenade (like the current M67, throwing lethal shrapnel) or a concussion grenade (like the old MK3A2, using blast overpressure to neutralize enemies in enclosed spaces) [47] [48]. This dual-mode means a soldier can carry one grenade type and decide on the fly which effect is needed, rather than carrying two different grenades [48]. Internally, it likely has a fragmentation sleeve or pre-frag matrix that can be enabled, or perhaps two chambers of explosive. The ET-MP's **electronic fuze** is a major leap: instead of a chemical delay igniter (which has variability of a few tenths of a second and cannot be easily adjusted), an electronic timer can be set with millisecond precision [49]. This allows very exact timing of the detonation (useful for synchronization or specialized tactics). Moreover, the detonation train is **"out-of-line"** until armed – meaning the detonator and main charge are physically separated by a barrier until the grenade is safely armed after throw [50] [49]. This greatly reduces risk of premature explosion; combined with modern *Insensitive Munitions* explosives, the ET-MP will be far safer to store, transport, and handle [51]. In fact, it's slated to be the first Army grenade to meet certain Insensitive Munition standards and to be approved by the Fuze Safety Review Board with an electronic fuze [52].

**Development History:** Research began around 2010, spurred by a request from warfighters for a more versatile grenade [53] [54]. By 2016, Picatinny engineers had prototypes and the program was publicly discussed. At that time, the plan was to transition ET-MP to the acquisition program office by Fiscal Year 2020 [49]. Indeed, an Army press release from 2016 stated *"the current plan for ET-MP is to transition to Project Manager Close Combat Systems in FY2020."* [49]. This implies that by 2020 they expected to have a design ready for final development, testing, and eventual production contracts. However, as of 2025, **ET-MP has not yet been fielded** to combat units. The Wikipedia entry (which needs updating) noted entry was expected in 2020 [55], but that schedule appears to have slipped.

**Current Status (2025):** The *exact* status isn't fully clear from public sources, but there are some indications. One clue is that the U.S. has in the meantime adopted or tested other grenades: notably the **Nammo Scalable Offensive Grenade** (a purely blast grenade with stackable sections) which by 2023 had completed first article testing and was being readied for fielding as the MK21 Mod 0 [56] [34]. It appears the Army prioritized replacing the offensive concussion grenade (which ET-MP would have covered in one mode) by directly buying Nammo's design. ET-MP, being an R&D project, may still be progressing slowly or have been temporarily deprioritized. There is no announcement yet that troops have received ET-MP grenades for operational use. It likely remains in **prototype or testing phase**, possibly facing technical challenges (electronic fuzes in small packages must be extremely rugged and reliable). It's also possible the program was folded into a larger effort or awaiting further funding.

That said, testing events probably occurred in 2018–2019 for safety and performance. Key aspects like ensuring the electronic fuze works in all environments (hot, cold, after being jostled) and passes safety certification would have been major hurdles. The concept itself is proven in artillery and munitions (electronic fuzes exist in shells), but applying it to a hand grenade – which is a **throwable, 5-second fuze device – is novel**. We do know that prototypes existed and were being demonstrated internally. By late 2023, there's still mention of ET-MP in literature, but without a fielding date, suggesting it's not yet in soldiers' hands.

**Timeline for Deployment:** Since FY2020 was missed, one might speculate deployment around mid-2020s if the project continued steadily. It could potentially enter service by, say, 2026–2027 if testing finishes and production is funded. However, absent official updates, this is uncertain. The Army might also be evaluating if the combination of M67 + MK21 (Nammo) covers enough capability without ET-MP. If ET-MP proceeds, expect a low-rate initial production, then fielding to select units (maybe special forces or infantry schools) to trial its effectiveness.

In summary, **ET-MP is at the cutting edge of grenade tech**: it embodies what future grenades will be like (smart, selectable effects, very safe until armed, precise timing) [49] [51]. Its development underscores how training must eventually adapt – for example, if soldiers have grenades with multiple modes, they must be trained to set the mode correctly under stress. That means **training grenades will need to simulate mode selection** too. One day, a training grenade might have a dummy switch to choose frag or blast, teaching troops to make that decision. For now, ET-MP itself is not fielded, but once it is, the training community will likely produce an inert or practice ET-MP (perhaps a variant of the Nammo training grenade or a new design) that allows practicing flipping that switch and feeling the differences in weight or employment.

**Tests and Trials:** To date, what's been done includes design tests, safety tests, and likely some user evaluations at Infantry School. No combat unit has used ET-MP in the field yet. When it's closer to deployment, the Army will announce it via Program Executive Office updates. Until then, **its timeline remains tentative**. The important takeaway is that ET-MP's concepts (electronic, dual-mode) will shape future requirements for both **combat grenades and their training simulators**.

## Other Notable Training Grenades and Simulators (Competitors and Specialized Models)

Beyond the devices listed above, there are several other **training grenade products or systems** in use or development across militaries and police forces. These range from alternate smart grenade simulators to devices tailored for **counter-terrorism (CT)** training or law enforcement needs. Below we highlight a few:

- **Saab Hand Grenade Simulator (HGS):** Saab Training Systems (Sweden) has long provided live training solutions (often paired with laser engagement systems). They have a **Hand Grenade Simulator** (HGS or HGS II) which is essentially a dummy grenade that, when "thrown" and triggered, will send out a coded laser pulse to simulate the grenade's effect in a MILES-like manner. Earlier versions included **fragmentation simulators and stun (flashbang) simulators** that could be selected by using different cartridges [57]. These devices didn't produce a loud bang or flash themselves (some might make a small pop), but they would cause MILES gear on targets to register a kill within a certain radius. Saab's simulators have been used by various NATO armies in combat training centers. While effective for scoring, they lacked the sensory realism that newer devices like Sim Grenade or FBSS now provide. Saab continues to integrate newer pyrotechnics (like Wescom's) and electronic components to improve their systems. Their

latest offerings tie into comprehensive instrumented ranges for **urban combat training**, where all weapons effects (rifle, AT4, grenades) are simulated and tracked.

- **M69 Practice Grenade (US):** The U.S. military's standard practice grenade for decades has been the **M69**, a blue-painted metal grenade that uses the M228 training fuze. It functions somewhat like the Nammo grenade: after a 4-5 second delay, it produces a small **pyrotechnic report (bang)**. The body can theoretically be reused (it's basically a cast iron hull), but usually these are one-time use in practice (the body may split or the cost of refurbishing isn't worth it). The M69 has been in service for basic training; its cost is relatively low (historically around 30-40 USD each) [58]. However, because it scatters small metal pieces from the fuze and emits smoke, ranges need to be cleared of debris. Also, its **casualty radius simulation is crude** – anyone within about 5 meters is considered "hit" in training, often enforced by instructor rulings. The M69 is essentially a predecessor to devices like Nammo's, but Nammo improved upon it with reusability and better safety certification.

- **Other** Reusable Grenade Simulators for Police/CT***: In law enforcement and CT training, scenarios often focus on* *stun grenades (flashbangs)** rather than frag grenades. Several companies have innovated here:

- *ALS TRMR-LE*: A **reusable flashbang device** that uses a 12-gauge blank shotgun primer or cartridge. It looks like a small metal canister; after "pin" removal, upon impact with the floor it detonates the blank, producing a loud bang (~130 dB). It can be reloaded with new blanks and used again. Police agencies like it for repeated room entry training – it's cost-effective and safer than actual flashbangs (no significant blast, just noise).
- *CTS (Combined Tactical Systems) Training Bang*: CTS produces the flashbangs used by many police/military. They offer **training versions** that output a reduced bang (e.g. 120 dB) and produce no shrapnel. Some are **reloadable bodies with screw-in charges** [59]. These allow practices in hallways or close quarters with less risk of hearing damage or fire, while still giving a stun effect.
- *Electronic Flashbang Simulators*: Devices like the **FoxFury T.E.D.D. (Tactical Electronic Distraction Device)** are non-pyrotechnic alternatives that use **bright LED strobe lights (up to thousands of lumens) and a loud siren** to replicate a flashbang's effect [60] [61]. They are reusable and programmable (some can do multiple "bangs" or strobes in sequence). Their advantage is no explosive at all – legal in places where pyro isn't. However, their sound may not be as overwhelming (often around 130 dB) and the light, while disorienting, is not exactly the same as a flash. These are used in training where safety is paramount or in scenarios where you cannot risk any fire (e.g. aboard aircraft for hijack drills).

- *Liberty Dynamic Enhanced Diversionary Device (EDD)*: A new product which is a **hybrid flashbang** – it uses a **binary reactive compound and a digital timer** to produce a flashbang effect with less flame and no ejected fragments. It's safer to use (low fire risk) and also has **zero delay variance** due to electronic timing. Some police and military units have tested it as a safer alternative to classic M84 flashbangs [62]. This shows the move towards **smart grenades in the stun category too**.

- **Airsoft and Simulation Grenades:** In the commercial and milsim world, companies like TAGinn (Russia) and Enola Gaye (UK) produce grenades (like the **TAG-67** and **EG-67**) that actually throw out BBs or fragments of dried peas, mainly for airsoft games. Interestingly, some military and police use these for force-on-force training because they provide a harmless projectile effect (marking or stinging the target to signify a "hit"). The **TAG-67**, for example, is a CO2-powered or pyro grenade that on detonation spreads a small burst of BBs over a 5m radius – enough to let players know they were "hit." It's 95% biodegradable (paper shell and plaster pellets) [63]. While

not standard issue for armies, they can be useful for certain training drills (e.g. riot control training or scenario-based training with non-lethal engagement, where the goal is to have the trainees feel something when the grenade goes off). These are very low-power and safe with eye protection.

- **Competing Smart Grenade Systems:** Serious Simulations is not alone; other firms have been pursuing similar concepts. There are patents (e.g., by Cole Engineering or other defense contractors) for **laser or RF-based grenade simulators** [64]. The U.S. Army's STE effort had multiple vendors; Serious Sim won the contract, but likely others (like Cole, Cubic, etc.) had prototype smart grenades. It's possible in the future we'll see alternate products that use slightly different methods (like perhaps an **ultrasonic ranging grenade** or one that deploys a sensor on impact).

- **Specialized CT Grenades:** For counter-terrorism units, one interesting development is **micro-pyrotechnic devices for breaching simulators or confined spaces**. E.g., small flash charges on a **"pyro card"** can simulate multiple effects (gunshots, grenade blasts) in indoor kill houses with minimal smoke [65] [66]. These aren't exactly grenades you throw, but rather fixed simulators that add battlefield effects. Still, they compete in the space of creating realistic training environments.

In conclusion, the landscape of training grenades includes everything from **high-fidelity digital systems** to **simpler reusable flashbangs and inert grenades**. The main competitors and alternatives not originally listed can be summarized by their focus: - **Laser/MILES-based grenades** (e.g., Saab HGS) – good for scoring hits but lacking noise/flash. - **Pyrotechnic flash-bang grenades** (e.g., FBSS-Mini, CTS, ALS devices) – great for sensory effect, one-time use, often no automatic scoring. - **Reusable blank-fired grenades** (e.g., TRMR, Pavashot NPDD) – moderate noise, reusable, popular with police. - **Airsoft/paint training grenades** – mark targets or provide minor impact, used in non-military training or force-on-force with projectiles. - **Electronic grenades** (e.g., Sim Grenade, or emerging products) – integrate tech for realism and feedback, but usually cost more.

Each of these finds its niche depending on the training objective (be it marksmanship with grenades, stress inoculation, safe mass training, or full mission simulation).

## Cost Considerations: Reusable vs. Single-Use Training Grenades

When deciding on training grenade solutions, cost – both initial and long-term operating cost – is a crucial factor. Different types have different cost profiles:

- **Reusable High-Tech Simulators (e.g., Sim Grenade):** These have a **high upfront cost** per unit (they contain electronics, sensors, battery, etc.) but **low marginal cost per use**. Once you buy them, you can use them hundreds of times with minimal consumables (perhaps just battery charging or replacement parts occasionally). However, they may require maintenance (calibrating sensors, replacing padding, software updates), which is a cost in terms of labor and support. If one device can replace hundreds of single-use grenades, over time it might justify its cost. For instance, even if a Sim Grenade cost several thousand USD each, if it saves the purchase of 500 pyro grenades over its life, it could break even. One must also factor infrastructure: they rely on having URUs/MILES gear. So initial investment includes the whole system. In military budgeting, these costs are often justified for **combat training centers or large units**, but not for every platoon in daily training.

- **Reusable Body/Replaceable Fuze grenades (Nammo-style):** These have a **moderate upfront cost** (the machined metal bodies and a set of fuzes). The ongoing cost is the **consumable training fuzes**. Each throw "uses up" one training fuze, which likely costs on the order of tens of dollars (since it has pyrotechnics and small parts). Still, this is cheaper than a full pyrotechnic grenade because the expensive part (the body) is reused. Over time, the **cost per throw** is relatively low. Nammo advertises theirs as a **low-cost option** for safe handling training [25]. If, for example, a unit buys 100 grenade bodies and 1000 fuzes, they can conduct 1000 throws. The bodies might last many cycles; only if a body is damaged does it need replacement. In contrast, if using one-piece simulators, you'd buy 1000 grenades for 1000 throws. Thus, long-term, the Nammo approach is more economical for large volumes of training. Many NATO armies try to reuse where possible for cost – e.g., some practice grenades allow refilling or new blanks.

- **One-Time Pyrotechnic Grenades (FBSS-Mini, M69, etc.):** These are **consumed with each use**, so the cost is linear with number of throws. The unit cost can vary: simple ones like M69 might be relatively cheap (perhaps $20-$50 each in bulk [58]), whereas sophisticated ones like FBSS-Mini cartridges could also be in that range or higher. Using 1000 of them means buying 1000 of them; there's no reuse. However, initial costs are lower – you don't need to buy an expensive device or infrastructure; you just buy as needed. For short-term or limited training (or armies with low training budgets), this pay-as-you-go model might be easier to manage. But for **sustained training programs, costs accumulate**. Additionally, each one-time pyro has to be disposed/cleaned up, and you have hazardous shipping/storage costs (they are explosives class items).

- **Maintenance and Replacement:** Reusable devices might incur repair costs if they break. Electronics can fail due to rough handling; padding can wear out. Those are additional costs that one-time grenades don't have (they either function or you discard them).

- **Comparative Example:** Imagine training 500 soldiers, each throwing 2 grenades in training. That's 1000 grenade uses.

- If using **M69 practice grenades**: need ~1000 units. If $30 each, that's $30,000 for that event (plus range cleanup costs, which might be minor).
- If using **Nammo grenades**: You could buy, say, 100 bodies (maybe ~$100 each, just a guess) = $10,000, and 1000 fuzes (maybe $10 each?) = $10,000. Total ~$20,000. After training, the 100 bodies remain for future use (they might last years). Next time you only buy more fuzes. So initial cost similar or slightly lower, and subsequent training events cheaper.

- If using **Sim Grenade**: Maybe you buy 50 Sim Grenades at, say, $2,000 each = $100,000 upfront. And you need the URU/MILES gear (cost not trivial, possibly another few hundred thousand for a whole unit's receivers). That's a big capital expense. However, once owned, you can conduct countless training exercises with minimal additional cost. Over a span of years, it could be economical if heavily utilized (like CTCs using them daily). But for a one-off training of 1000 throws, it's not cost-efficient. So these are justified when integrated into **frequent training cycles** or for high-value training (like prepping a brigade for deployment with a very realistic exercise).

- **Flashbang simulators vs. live flashbangs (for police):** Live flashbangs can cost $50-$100 each and you obviously can't reuse them. Training simulators like the TRMR reusable might cost a few hundred upfront but then only pennies per blank thereafter. Departments that train often save money with reusable flashbang trainers. Similarly, electronic ones are pricey ($800+ each for something like a T.E.D.D.) but can be used thousands of times (only battery changes).

- **Environmental and logistical cost:** One-time use grenades might incur environmental mitigation costs. For instance, picking up debris (especially if not biodegradable) costs labor, and some chemical compounds (like older simulators with perchlorates) might require environmental cleanup. The Army has even modified some simulators to be perchlorate-free to save such costs [67].

In general, **reusable systems have higher initial procurement and require a long-term approach to realize savings**, whereas **single-use systems are cheaper to start but cost more cumulatively**. Militaries with steady budgets and long-term outlook (and emphasis on sustainability) are trending toward reusable or hybrid solutions (e.g., use electronics where possible, use one-time pyro only where needed). However, it's often practical to have a mix: reusable for everyday drills and one-time pyros for special realism injections.

Additionally, one must consider **training value per cost**. A Sim Grenade might be expensive, but if it substantially improves training outcomes (more realistic, measurable, and thus better prepared soldiers), that value might justify its cost in ways that a purely financial comparison doesn't show. Conversely, if a unit cannot afford enough Sim Grenades for everyone, cheaper grenades at least get more hands-on practice distributed among troops.

**Summary:** - **Nammo-style reusable inert grenades**: moderate cost, mid-range realism (sound only), good for bulk use – likely the best bang for buck for basic training (most **cost-effective** for handling training) [25]. - **High-tech smart grenades**: high cost but unique capability (scoring, realism). Cost-effective mainly for high-end use (it's an **investment in quality of training**). - **Disposable pyrotechnic grenades**: lower unit cost, scalable purchase (you buy as needed). Good for occasional training or adding realism, but **most expensive over large numbers** and have hidden costs (safety gear, cleanup).

An actual cost comparison study would gather real procurement prices, but since those are rarely public, the above is a reasoned qualitative comparison. In practice, many militaries use a **tiered approach** to balance cost: e.g., use lots of cheap dummies for practice throws, a fair number of reusable fuzes for more realism, a few pyrotechnic simulators for stress effect, and maybe a handful of high-tech devices for complex exercises. This way, they allocate budget where it yields the most training value and ensure sustainability.

## Pros and Cons of Each Type in Specific Training Scenarios

Finally, we evaluate the advantages and limitations of each of the discussed training grenade models in different **specialized training scenarios: force-on-force combat exercises**, **urban combat/CQB training**, and **basic safe handling training** (no-injury training).

### 1. Serious Simulations' Sim Grenade™

- **Force-on-Force:** *Pros:* Excellent for force-on-force. It provides **real-time hit adjudication** – no arguments whether someone was "within the blast," as the system tells you [1] [2]. It also accounts for cover, adding tactical depth (players learn to use cover properly) [7] [10]. Its reusability means it can be thrown in numerous engagements in a single exercise without logistics resupply. *Cons:* Requires that all players wear the URU/MILES gear and that the exercise is instrumented [16] [18]. This can be cumbersome and limit use to well-resourced exercises. Also, while it makes a bang/flash, the sensory effect is modest – players might hear it and see a flash, but it's not deafening or stunning (by design, to avoid injury). So in a force-on-force firefight, some might argue it doesn't *feel* as frightening as a real grenade. Additionally, the Sim Grenade's

electronics could theoretically glitch (false negatives/positives), so backup adjudication by observers is wise. But overall, it's the **gold standard for force-on-force grenade simulation** because of its scoring and realism in employment (throw like a real grenade).

- **CQB/MOUT:** *Pros:* It can be used in urban combat training to practice proper grenade throwing into rooms, because it's safe to use around role-players (padded exterior, no fragments) [13]. It will register casualties in adjacent rooms if within radius, teaching trainees about the importance of distance and cover in buildings. It's also good for **AAR in MOUT** – you can reconstruct who got "hit" in a houseclearing and why. *Cons:* In tight CQB environments, one major missing piece is the **disorientation effect**. The Sim Grenade's bang might not be loud enough to force trainees to react as they would to a real stun grenade. Thus, enemies (role-players) might not be as suppressed by it, unless they play along. Therefore, for elite CQB units, Sim Grenade might be seen as too "polite" – they often prefer flashbangs (or at least the FBSS-Mini) to train the fight-through-the-blast scenario. Another con: indoor use of the Sim Grenade might be limited by the need for line-of-sight signals; if heavy walls block signals, someone in the next room might not get tagged even if in real life they'd hear the explosion – however, the device is tuned to approximate real blast radius through walls [7], so that's partly mitigated by design.

- **Safety Training (no-injury focus):** *Pros:* The Sim Grenade is built to be **very safe** – no actual blast, so there's no injury risk from fragments or pressure. It can even hit a person when thrown and not injure them (thanks to padding) [5]. This makes it safer than using something like a hard metal dummy. So it could conceivably be used for some handling training as well, with minimal risk (except maybe dropping the device on one's foot – it's about the weight of a real grenade ~14 oz). *Cons:* It's arguably overkill for basic training. For a new soldier just learning how to throw, the high-tech features aren't needed, and the cost wouldn't justify issuing these for every recruit on day one. Also, one might worry a recruit could misuse or accidentally break an expensive item. So Sim Grenade is not typically used for entry-level training; it's for more advanced exercises. In terms of maintenance safety, it has electronics and batteries – minimal hazards, but instructors have to ensure it's charged and functioning (a dead battery mid-training could be a minor hiccup).

**Summary:** Sim Grenade **excels in realistic combat simulation and objective feedback**, making it top choice for complex force-on-force field exercises. It is less effective for sensory shock training in CQB (due to limited bang) and generally not economical for basic throw training. Its strong points are realism and scoring; its weak points are cost, infrastructure requirements, and lack of intense flashbang effect.

## 2. Nammo Training Hand Grenade (Reusable Inert Grenade)

- **Force-on-Force:** *Pros:* It is somewhat useful in force-on-force in that it can be thrown and will make a noise to signal detonation. Troops can agree that anyone within a certain radius or in the same room is "hit." It's also safe enough to throw *toward* opposing players (it's inert except a small pop). *Cons:* **No automatic hit detection** – reliance on exercise umpires or honor system. This can lead to disputes or inconsistent outcomes. Also, its bang is not loud enough to ensure everyone knows it went off in a chaotic battle; people might not notice it amidst gunfire. In large outdoor force-on-force, a small pop is easily missed. So while it's fine for **structured scenarios (e.g., section attack where an umpire says "grenade went off, these guys are down")**, it's not ideal for free-form exercises. In most modern force-on-force (with MILES), a Nammo grenade would not interface unless someone jury-rigs a MILES squib to it.

- **CQB/MOUT:** *Pros:* The Nammo grenade is **good for teaching soldiers how to employ grenades in urban settings** – e.g., how to safely cook-off (if allowed) and throw into a room and then clear the room after the bang. The modest report provides a cue to enter. Because it doesn't produce a dangerous blast, it can be used in shoot houses repeatedly. *Cons:* It doesn't actually stun anyone. So for CQB where role-players or targets are expected to be affected by a grenade, again it has to be scripted (targets pretend to be incapacitated when they hear the pop). Many CQB instructors prefer using at least a flashbang simulator to train the real effects; a quiet inert grenade might build bad habit of underestimating grenades. Also, in tight spaces, even a small pop can be loud, but the Nammo's pop is far less than a real flashbang – it won't trigger the full adrenaline/fear response.

However, for **tactical movement training in urban areas** (like how to approach a door, toss a grenade in, then enter), Nammo grenades are very suitable and commonly used because they allow repeated practice without damaging the facility. They are often used in pairs with blank firing: soldiers throw the training grenade (pop), then assault the room firing blanks – giving a somewhat realistic battle noise scenario without any real explosives.

- **Safety Training:** *Pros:* This is the **core strength** of the Nammo grenade. It's perfect for teaching basic grenade handling, as mentioned. All soldiers can practice pulling pins and throwing with a delay pop, as many times as needed. It's **safe enough for regular training** (approved by Finnish Army for even small ranges) [28] [30]. You can throw it, pick it up, do it again – fostering confidence. *Cons:* There's still a minor risk: the fuze does have a small explosive charge, so if a soldier were to improperly hold it (not throwing it, letting it pop in hand), there's a risk of minor burns or injury. So basic safety must still be observed (but that's part of training!). Also, while much safer than live, it's not as totally benign as the THG67 – i.e., you wouldn't purposely let it explode right next to someone's head.

Nammo grenades cost money for each fuze, so for budget reasons you might not let recruits throw 50 of them each – they might get to throw a handful in training. But since they can be reused, each soldier could get multiple repetitions which is good for learning.

**Summary:** Nammo's reusable grenades are **jack-of-all-trades for training**: **very good for basic training**, decent for rehearsing battle drills in urban or field environments, but **not effective for scoring or strong blast effect**. Pros: realistic feel, moderate bang, reusability, safety certified. Cons: no built-in simulation of casualty effect, limited sensory impact. They fill the middle ground between pure dummy and full simulator.

## 3. Wescom FBSS-Mini (Flash-Bang-Smoke simulator)

- **Force-on-Force:** *Pros:* Using FBSS-Mini in force-on-force can add **realism and stress**. For example, during an exercise, an observer/controller could fire off an FBSS-Mini where a grenade was thrown to make everyone react to the loud bang – increasing **psychological fidelity** of the simulation (soldiers will duck, communication may be momentarily impaired, etc.). This can be integrated with a MILES system by having, say, a paired laser simulator. The **immediate benefit** is players cannot ignore grenades; the sound forces them to acknowledge "grenade went off." *Cons:* Alone, FBSS-Mini doesn't discriminate friend/foe casualties. It's purely an effect; umpires still have to enforce who is dead. Also, frequent use in a force-on-force might be problematic: multiple huge bangs can confuse the exercise or cause hearing issues if protection is insufficient. For large exercises, logistics of issuing dozens of pyro charges and ensuring none cause fire in dry terrain is a con. Typically, pyro simulators are used sparingly (like artillery sims at certain points). Overuse can also desensitize soldiers or overshadow other training points.

Another con is safety in a free-form force-on-force: if an FBSS-Mini grenade lands very near someone, even though non-lethal, it can cause temporary disorientation or minor burns. There's a small chance of someone getting a tinnitus or a bit of debris in the eye if not well protected. Thus, many armies restrict their use to more controlled environments (like MOUT villages or specific lanes).

- **CQB/MOUT:** *Pros:* This is where the FBSS-Mini shines. In urban breaching and room clearing drills, it provides the **closest experience to a real grenade or flashbang**. When the device goes off, the entry team experiences the loud crack and bright flash, training them to keep moving through it (or to capitalize on it if they threw it). The opposing role-players get realistically stunned (if they aren't wearing as much ear protection – though usually everyone in training has some ear pro, real bad guys wouldn't). It's great for **battle inoculation** – after a few times, soldiers learn to keep focus even with intense noise. *Cons:* As mentioned, ear protection dulls the effect a bit, but you can't trade safety for effect too much. Also, indoor use means a lot of smoke build-up – after a few grenades, visibility can drop (could be a pro or con; some trainers like it to simulate battlefield smoke, others pause training to ventilate). Another limitation: cost and preparation – you have to reload grenades with these cartridges each time, which takes some time and care. If a scenario needs multiple grenade throws in quick succession, each soldier needs multiple grenades or a quick way to reload between runs.

In **hostage rescue or CT scenarios**, often teams will throw multiple flashbangs – using training simulators like FBSS allows them to do that realistically. For instance, a counter-terror unit storming a mock aircraft might throw 6 flashbang sims in different sections; the chaos of noise and flash is essential to train controlling adrenaline and coordinating under stress.

- **Safety Training:** *Pros:* FBSS-Mini is not used for basic safe handling training – it's simply too loud and overkill. However, one could argue it contributes to safety by getting soldiers used to explosions. *Cons:* If used with unseasoned recruits without proper acclimatization, it could cause panic or even physical issues (like ruptured eardrums if they weren't wearing protection right). So these are reserved for troops who have already mastered basic grenade handling and are moving into tactical application with stress.

One area of "safety" it does address is **training to not fear your own grenade**. Sometimes soldiers flinch or don't throw far enough due to fear of the bang. By training with loud simulators, they become confident to throw grenades properly (not just drop them from cover and duck too early).

**Summary:** The FBSS-Mini's **advantages** lie in **sensory realism** – it's unmatched in delivering the psychological effects of a grenade blast during training [35]. This makes it indispensable for CQB, urban combat, and stress-inoculation training for both military and police. Its disadvantages are that it **lacks built-in scoring, is one-time use, and requires safety measures** (hearing protection, fire precautions). It's not for every training day; it's for those scenarios where you intentionally push trainees' sensory overload to build resilience and proper reactions under fire.

## 4. SOHNI THG67 (Biodegradable Low-Impact Grenade)

- **Force-on-Force:** *Pros:* Essentially none in terms of direct simulation – a THG67 or similar is not meant to simulate casualty-causing grenades in an opposed exercise. If one side throws it, the other side might not even realize it was a "grenade" if the effect is tiny. It could perhaps be used as a proxy just to indicate that a grenade was thrown (like a judge sees it land and calls kills). But it offers no loud signal or laser, so it's purely an honor-system dummy. *Cons:* Using such an inert grenade in force-on-force can lead to confusion or disagreement ("did it go off? were you in range?"). Therefore, they're generally *not used in force-on-force* beyond maybe a very controlled

scenario where safety is so paramount that no pyro is allowed (in which case, one might as well not simulate grenades at all, or simply call "grenade" vocally – which some exercises do).

However, one could conceive a scenario: e.g., a cadet exercise in protected forest where they use biodegradable grenades so as not to pollute – they throw them, and observers decide casualties. This would prioritize safety and environment over realism, likely only done in low-level training.

- **CQB/MOUT:** *Pros:* For initial urban training – like learning how to clear rooms *with* grenades conceptually – these can be used. Trainers can have soldiers practice the sequence: cook grenade, yell "frag out," toss into room, then enter after a "pop." Since it won't injure anyone, multiple people can stand nearby, making it good for walk-throughs and slow rehearsals. Also, since it breaks into non-cutting pieces, you can use it in shoothouses without damaging the facility or leaving metal fragments. *Cons:* In true CQB dynamic training, it provides no stun effect and thus is not really used. Once teams are doing live-fire or high-intensity clears, they either go with blanks + flash sims or nothing. A purely inert grenade might actually become a hazard if someone doesn't notice it and slips on the pieces (though biodegradable bits like clay are usually just powder after explosion, so probably fine).

Where they do help is in **scenario-based training where environmental impact is a concern**. For example, a unit practicing on a heritage site or public urban area (as occasionally done for drills) could use these so as not to litter or cause damage while still practicing throwing grenades (with permission, of course).

- **Safety Training:** *Pros:* This is the **forte** of THG67-type grenades. For teaching rookies grenade safety, it's unbeatable. A large group of recruits can train simultaneously throwing in an open field without fear – even if one goes off too close, nobody is seriously hurt [43]. It's great for teaching the **throwing technique**: because trainees are often very nervous about live grenades, starting with something that won't kill them even if they fumble helps build proper form. Also, since it's biodegradable, you don't need to waste time policing brass/chips – good for throughput on ranges. *Cons:* The downside might be that if a trainee only experiences this very tame grenade, they might underappreciate a real grenade's power. So usually after initial confidence-building with these, the training escalates to louder simulators or eventually live. Another con: these grenades might not be as readily available in all armies (they are somewhat niche products), so sometimes units improvise (like using practice grenades with reduced charges similarly).

**Summary:** The THG67 and similar grenades are **excellent for safe, environmentally friendly training of basic grenade handling and large-scale exercises where cleanup or safety is an issue** [43]. They have **no role in realism-focused scenarios** like force-on-force or stress shoots, because their effect is minimal. Pros: *ultimate safety*, *no debris*, *good weight replication*. Cons: *minimal sensory effect*, *no scoring*, *not suitable for simulated combat*. They are a foundational tool for initial training but are phased out as training progresses to more advanced levels.

## 5. ET-MP (Future live grenade and its training implications)

*(While ET-MP is not a training grenade, a brief note on how its features influence training scenarios is warranted.)*

- **Force-on-Force:** Once ET-MP is fielded, force-on-force training would need simulators that replicate its dual-mode function. A soldier in an exercise might have to indicate whether they're "throwing frag or blast." This could be done via a setting on a Sim Grenade (future version) or simply by communicating. The complexity here is ensuring the training device can emulate both

a fragmentation kill radius and a concussion effect (perhaps a smaller radius but through walls). It's likely that training systems like STE will incorporate ET-MP parameters. The timeline of ET-MP deployment will dictate when such simulators are needed. But conceptually: *Pros:* ET-MP in real use gives troops flexibility, and in training that flexibility must be tested. *Cons:* Until simulators catch up, troops might have a live capability they can't fully practice in force-on-force (except via abstractions).

- **CQB/MOUT:** ET-MP's blast mode is specifically for enclosed spaces – in training, soldiers will need to practice selecting that mode for bunker clearing or room clearing. Instructors will want to verify that they flip the switch to the correct mode before throwing (a new task under stress). So training grenades will need dummy mode switches. If not, instructors may simulate it by saying "imagine you set it to blast." *Pros:* Teaches decision-making – e.g., don't use frag mode in a hostage scenario, use concussion to avoid frags injuring hostages. *Cons:* It adds a small step that could be fumbled; training needs to iron that out, which means more repetition (cost in time and devices).

- **Safety Training:** Since ET-MP uses an electronic fuze that presumably can be made inert for training versions, the handling might be similar to current grenades. One notable safety difference: ambidextrous design – trainers must now ensure both left- and right-handed soldiers use the same procedure (which simplifies training, actually). Also, the electronic fuze might give an exact 5.0 second delay every time, versus current fuzes which have variability (sometimes blowing at 4 or 6 seconds). That means in training, soldiers might have to un-learn the old caution of "fuzes can vary" – which could be a pro (consistency) or a con (if they ever use older nades, they might be caught off guard by inconsistency).

**Summary:** ET-MP's eventual fielding will push training systems to adapt: **smart training grenades that simulate mode selection and electronic timing**. It underscores a broader trend: as operational grenades get "smarter," training grenades will also need to get smarter (with electronics) to provide appropriate muscle memory and decision training. This likely means the cost and complexity of training grenades will increase, but so will their fidelity to real weapons.

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In conclusion, each training grenade type has a clear role and set of strengths: - For **Force-on-Force** engagements, the **Sim Grenade** is unparalleled in fairness and data capture [1] [2], while **pyro simulators (FBSS)** can supplement realism at the cost of automated scoring [35]. - In **Urban/CQB training**, **flash-bang simulators (FBSS, etc.)** are critical for realism [35], though a **Sim Grenade** can still be useful for practicing tactical employment without actual casualties. Often, a combination is ideal (use Sim Grenade for mechanics and feedback, occasionally use a flashbang simulator to train reaction to effects). - For **Safe handling and basic training**, the **Nammo reusable grenade** and **biodegradable grenades (THG67)** are winners [25] [43]. They allow extensive practice with minimal risk and cost in the long run. - And looking ahead, as technology like **ET-MP** becomes reality, training will evolve to incorporate those features – ensuring that when soldiers finally pull the pin on a real multi-purpose grenade, it feels like second nature due to the comprehensive training with its simulated counterparts.

Each tool has its drawbacks (be it cost, lack of sensory effect, or lack of scoring), so trainers often employ them in **complementary ways**. By doing so, they can cover the full spectrum: cognitive decision-making (who to target, when to throw), physical technique (how to throw safely and accurately), physiological conditioning (not flinching at bangs), and tactical execution (clearing a room, using cover, etc.). The combination of devices like Sim Grenade and FBSS-Mini – along with traditional

practice grenades – ensures that modern soldiers and law enforcement officers receive **holistic grenade training** that is safe, cost-effective, and as close to the real fight as possible.

**Sources:**


1. UCF Business Incubation Program – Press release on Serious Simulations "Sim Grenade" [1] [4]
2. Orlando Tech News – Article on Army contract for simulated grenades [2] [15]
3. PEO STRI (U.S. Army) – Grenade & Claymore Training System description [16] [18]
4. Nammo AS – Official product page for Training Hand Grenade [25] [30]
5. Shephard Media – *Grenade programme highlights demand for training device* (Scott Gourley, 2024) [27]
6. Pyrotechnic Stores Handbook (Scribd) – Details on Wescom Defence FBSS-Mini [35] [37]
7. Army Technology – *Eco-Friendly, Non-Metallic Practice Hand Grenade* (whitepaper by Bristol Trust) [43]
8. Army.mil – Picatinny Arsenal news article on ET-MP grenade development [49] [48]


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[1] [3] [4] [7] [8] [11] [41] Serious Simulations Launches Production of First Ever "Sim Grenade" - UCF Business Incubation Program - University of Central Florida
https://incubator.ucf.edu/serious-simulations-launches-production-of-first-ever-sim-grenade/

[2] [6] [10] [12] [13] [14] [15] [23] [24] Orlando firm lands U.S. Army simulated grenades contract
https://orlandotechnews.com/2025/08/orlando-defense-firm-lands-u-s-army-contract-for-simulated-grenades/

[5] [9] [21] [22] Serious Simulations Awarded Prime OTA Contract for U.S. Army Synthetic Training Environment Live Training System (Simulated Hand Grenades & Mines) - UCF Business Incubation Program - University of Central Florida
https://incubator.ucf.edu/serious-simulations-awarded-prime-ota-contract-for-u-s-army-synthetic-training-environment-live-training-system-simulated-hand-grenades-mines/

[16] [17] [18] [19] [20] Grenade and Claymore Training System
https://www.peostri.army.mil/Project-Offices/PM-TRADE/PdM-STE-LTS/Rapid-Fielding-Grenade-Claymore/

[25] [26] [28] [29] [30] Training Hand Grenade - Nammo | Nammo
https://www.nammo.com/product/our-products/grenades-warheads-energetics/training-hand-grenade/

[27] [32] Grenade programme highlights demand for training device | Shephard
https://www.shephardmedia.com/news/training-simulation/grenade-programme-highlights-demand-for-training-device/

[31] nammo.com
https://www.nammo.com/wp-content/uploads/2023/09/Nammo-Handbook-2023.pdf

[33] [56] National Defense Digital September 2023 Page-8
https://digital.nationaldefensemagazine.org/september-2023/page-8?p=10

[34] [PDF] *For HQ POC, see "Deputy Program Manager List" on NAVSEA ...
https://www.navsea.navy.mil/Portals/103/Documents/Small_Business_Forum/r1_Enterprise_LRAF_03052020.pdf

[35] [36] [37] [38] [39] [40] [65] [66] Pyrotechnic Stores Handbook | PDF | Shell (Projectile) | Lighting
https://www.scribd.com/document/610632552/Pyrotechnic-stores-handbook

[42] Simulated Hand Grenade Having a Multiple Integrated Laser ...
https://patents.google.com/patent/US20100058947A1/en

[43] Eco-Friendly, Non-Metallic Practice Hand Grenade - Army Technology
https://www.army-technology.com/downloads/whitepapers/mines/eco-friendly-practice-grenade/

44 46 Training Ammunitions - 한국씨앤오테크(주)
https://en.cnotech.com/military1?seq=10

45 Bristol Trust - Army Technology
https://www.army-technology.com/contractors/training/bristol-trust/

47 48 49 52 53 Picatinny develops next generation of hand grenade | Article | The United States Army
https://www.army.mil/article/175189/picatinny_develops_next_generation_of_hand_grenade

50 51 54 55 Enhanced Tactical Multi-Purpose - Wikipedia
https://en.wikipedia.org/wiki/Enhanced_Tactical_Multi-Purpose

57 [PDF] OPERATOR MANUAL HAND GRENADE SIMULATOR (HGSII)
https://fcc.report/FCC-ID/R4AHGS2/3174069.pdf

58 4 Most Powerful U.S. Grenades in Military History - Thomasnet
https://www.thomasnet.com/insights/most-powerful-u-s-grenades/

59 ATT-Tactical™- CTS Training Flash-Bangs
http://www.att-tactical.com/att_cts_training-bangs.html

60 61 T.E.D.D. - Tactical Electronic Distraction Device | FoxFury
https://www.foxfury.com/t-e-d-d-tactical-electronic-distraction-device/?srsltid=AfmBOopYydWOzZ_qFK9kN9PRwvRL28cHZZbuJRjxhh5U58mj4cd6PhIY

62 Liberty Dynamic: The Safer Flashbang Alternative
https://www.libertydynamic.com/

63 Enola Gaye EG67 Airsoft Frag Training Grenade - MiR Tactical
https://mirtactical.com/airsoft/airsoft-grenades/enola-gaye-eg67-frag-training-grenade/

64 Training grenade for multiple integrated laser engagement system
https://patents.google.com/patent/US6065404A/en

67 Army Makes Training Grenades Safer for the Environment
https://www.army.mil/article/8581/army_makes_training_grenades_safer_for_the_environment